Nic Fishman

PhD student in Statistics, Harvard University

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Education

Harvard University — *Cambridge, MA*
Anticipated PhD in Statistics, advised by Prof. Marinka Zitnik — *Aug. 2023–Present*

University of Oxford — *Oxford, UK*
MSc in Statistics, advised by Prof. Yee Whye Teh — *Oct. 2021–June 2023*

Stanford University — *Stanford, CA*
B.S. in Computer Science and B.A. in Sociology with Honors – GPA: 3.9 — *Sept. 2017–June 2021*

Honors & Awards

2023	**James Mill Peirce Fellowship**, Harvard University	*Cambridge, MA*
2023	**Graduate Research Fellowship**, National Science Foundation	*Cambridge, MA*
2021	**Rhodes Scholarship**, Rhodes Trust	*Oxford, UK*
2021	**Outstanding Senior Thesis Award and Firestone Medal**, Stanford University	*Stanford, CA*
2020	**Spotlight Talk**, Computational Biology Workshop, ICML	*Vienna, AT*
2017	**G.R.E.A.T. Award**, NHGRI, National Institutes of Health	*Bethesda, MD*

Publications

Published and accepted

Count Bridges enable Modeling and Deconvolving Transcriptomic Data. **Nic Fishman**, Gokul Gowri, Tanush Kumar, Jiaqi Lu, Valentin De Bortoli, Jonathan S. Gootenberg, and Omar Abudayyeh. *International Conference on Learning Representations*, 2026.

Rethinking Repeatability in Observational Social Science. Jonathan Ben-Menachem, Ari Galper, and **Nic Fishman**. *Sociological Methods & Research (accepted)*, 2026.

Generative Distribution Embeddings. **Nic Fishman**, Gokul Gowri, Peng Yin, Jonathan Gootenberg, and Omar Abudayyeh. *Advances in Neural Information Processing Systems*, 2025.

Partisan disparities in the funding of science in the United States. Alexander C Furnas, **Nic Fishman**, Leah Rosenstiel, and Dashun Wang. *Science*, 2025.

The fragility of fairness: Causal sensitivity analysis for fair machine learning. Jake Fawkes, **Nic Fishman**, Mel Andrews, and Zachary C Lipton. *Advances in Neural Information Processing Systems*, 2024.

Estimating Vote Choice in U.S. Elections with Approximate Poisson-Binomial Logistic Regression. **Nic Fishman** and Evan Rosenman. *NeurIPS 2024 Workshop on Optimization for Machine Learning*, 2024.

Human mobility networks reveal increased segregation in large cities. Hamed Nilforoshan, Wenli Looi, Emma Pierson, Blanca Villanueva, **Nic Fishman**, Yiling Chen, John Sholar, Beth Redbird, David Grusky, and Jure Leskovec. *Nature*, 2023.

Diffusion models for constrained domains. **Nic Fishman**, Leo Klarner, Valentin De Bortoli, Emile Mathieu, and Michael Hutchinson. *Transactions on Machine Learning Research*, 2023.

Metropolis sampling for constrained diffusion models. **Nic Fishman**, Leo Klarner, Emile Mathieu, Michael Hutchinson, and Valentin De Bortoli. *Advances in Neural Information Processing Systems*, 2023.

Change we can believe in: Structural and content dynamics within belief networks. **Nic Fishman** and Nicholas T Davis. *American Journal of Political Science*, 2022.

Should attention be all we need? The epistemic and ethical implications of unification in machine learning. **Nic Fishman** and Leif Hancox-Li. *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*, 2022.

Systematic characterization of generative models for de novo design of regulatory DNA. **Nic Fishman**, Avanti Shrikumar, Georgi Marinov, and Anshul Kundaje. *ICML Computational Biology Workshop (Spotlight)*, 2020.

Preprints and working papers

Editorial Screening when Science is Cheap. **Nic Fishman** and Gabriel Sekeres. *Working paper*, 2026.

Distribution-Conditioned Transport. **Nic Fishman**, Gokul Gowri, Paolo LB Fischer, Marinka Zitnik, Omar Abudayyeh, and Jonathan Gootenberg. *arXiv preprint arXiv:2603.04736*, 2026.

Making (Global) Criminal Procedure: Empire, the End of Justice, and the Rise of Efficiency. **Nic Fishman**. *Stanford University*, 2021.

Presentations

2025	**Generative modeling on the space of empirical measures**, Boston Diffusion Day	*Cambridge, MA*
2025	**Generative modeling on the space of empirical measures**, Harvard Statistics Seminar	*Cambridge, MA*
2025	**False Promises and False Premises of Fair Machine Learning**, Columbia Algorithms and Society Seminar	*New York, NY*
2025	**Predicting perturbations at single-cell resolution**, Cell Circuits and Epigenomics Seminar	*Cambridge, MA*
2023	**Constrained Riemannian Diffusion Modeling**, Thesis Defense	*Oxford, UK*
2022	**Invited talk, Should attention be all we need?**, Stanford Human Centered Artificial Intelligence	*Stanford, CA*
2022	**Invited talk, Should attention be all we need?**, DeepMind Sociotechnical Systems	*London, England*
2022	**Making (Global) Criminal Procedure**, Law and Society Association Annual Meeting	*Lisbon, Portugal*
2021	**Structural and content dynamics within belief networks**, American Political Science Association Annual Meeting	*Seattle, WA*

Research Experience

Harvard Medical School, Harvard University — *Cambridge, MA*
PhD research with Prof. Marinka Zitnik — *Apr. 2024–Present*

- Building agentic systems to automate statistical method development, with applications in biology and healthcare.
- Developing multiscale models for biological systems spanning genomics, single-cell RNA-seq, and spatial transcriptomics, with applications to perturbation prediction.
- Developing EHR-based methods for treatment-effect estimation and policy optimization at health-system scale, including ongoing work with Epic on clinically interpretable decision support.

Harvard Medical School, Harvard University — *Cambridge, MA*
Research with Profs. Jonathan Gootenberg and Omar Abudayyeh — *May 2024–Present*

- Developing transport and deconvolution methods for single-cell perturbation modeling, enabling generalization across unseen perturbations and cellular contexts.
- Designing transformer architectures for set-valued representations of cell populations to support fine-grained *in silico* experimentation and response prediction.
- Led benchmarking and validation efforts for out-of-distribution prediction in single-cell sequencing, unifying existing methods and evaluation metrics.

Institute for Quantitative Social Science, Harvard University — *Cambridge, MA*
Research with Prof. Kosuke Imai — *June 2020–Sept. 2024*

- Constructed a non-parametric framework for causal inference under general interference using regression.
- Developed efficient algorithms and theory for non-parametric monotone regression for estimation and inference in the general interference framework, with applications in large-scale network and spatial experiments.
- Worked on adaptive methods for pure exploration in combinatorial designs in generalized linear models.

Department of Statistics, University of Oxford *Oxford, UK*
Research in OxCSML Group under Yee Whye Teh *Sept. 2021–June 2023*

- Built constrained diffusion and flow-matching models for cyclic peptides and antibody loops, connecting generative modeling to molecular design problems.
- Developed techniques for extending diffusion models to constrained domains in general Riemannian geometries.

Center for Science of Science & Innovation, Northwestern University *Evanston, IL*
Research Fellow *June 2021–Sept. 2023*

- Synthesized huge international datasets on scientific grantmaking and publication to generate a picture of the landscape and political implications of scientific funding.
- Leveraged LLMs to develop fine-grained topic classifications across grants and paper abstracts to understand when grant-making institutions lead scientific investigation as opposed to following what is already being studied.

Economics Department, Massachusetts Institute of Technology *Cambridge, MA*
Research Assistant for Prof. Victor Chernozhukov *June 2021–June 2022*

- Developed methods for dynamic treatment effect estimation and policy learning in healthcare.
- Specifically extended g-estimation to non-linear blip functions using tools from minimax optimization.

Stanford Center on Poverty and Inequality and Stanford Network Analysis Project *Stanford, CA*
Research Intern for Prof. Jure Leskovec and Prof. David Grusky *Jan. 2020–June 2021*

- Used path crossings to study socioeconomic stratification in the United States, developing insights about heterogeneity in segregation across different social contexts and across different income deciles.
- Worked on algorithms to identify when individuals cross paths in time and space using massive GPS data.

Kundaje Lab, Stanford AI Lab *Stanford, CA*
Research Intern for Prof. Anshul Kundaje *Jan. 2018–June 2021*

- Created modular system for generating DNA sequences using GANs/VAEs/Transformers and optimizing these models to produce sequences with specified properties (level of gene expression or chromatin accessibility).
- Developed a k-NN algorithm to evaluate (1) the fidelity of samples from generative models and (2) the robustness of neural network predictions on regression outputs (extending existing work on classification).

Work Experience

Knowledge Graph Team, Google *San Francisco, CA*
Research Intern *Summer 2022*

- Research integrating LLMs and knowledge graphs (KGs) for search, particularly combining LLM representations with graph neural networks over a knowledge graph to produce KG-informed representations.
- Developed methods to identify author expertise using LLM/KG representations for multimedia topic modeling.

Data for Progress *New York, NY*
Data Scientist *June 2018–May 2021*

- Led development of polling infrastructure that produced the most accurate poll results in the Democratic Primary. Built out MySQL database for storage and analysis of survey responses. Automated chart and report generation from this database. Developed search engine and website for internal use to assist in research.
- Designed, conducted, and analyzed polls used to guide policy change for the Green New Deal, Medicare for All, several HR-1 issues, and criminal justice reform, among other progressive issues.
- Developed novel poll weighting scheme achieving state-of-the-art accuracy.
- Automated argument detection so that non-technical colleagues to easily interpret open-ended survey responses in policy briefs by building hierarchical Dirichlet process models for non-parametric topic modeling.
- Created an ecological loss function and corresponding neural networks, extending ecological inference.
- Led team creating word2vec models to analyze gender/racial bias in news articles around the 2016 election.

Sunrise Movement *Washington, DC*
Data Science Consultant *Fall 2020*

- Used LASSO and random forest algorithms to build interpretable heuristics for identifying low-propensity swing state voters for a large-scale get-out-to-vote campaign in advance of the 2020 presidential election.

Star Lab Corporation *Washington, DC*
Machine Learning Specialist *Summer 2018*

- Worked on a Red Hat kernel module to log system activity and leveraged it for neural net anomaly detection.